MATH 420-01

2.3 Composition of Linear Transformations and Matrix Multiplication

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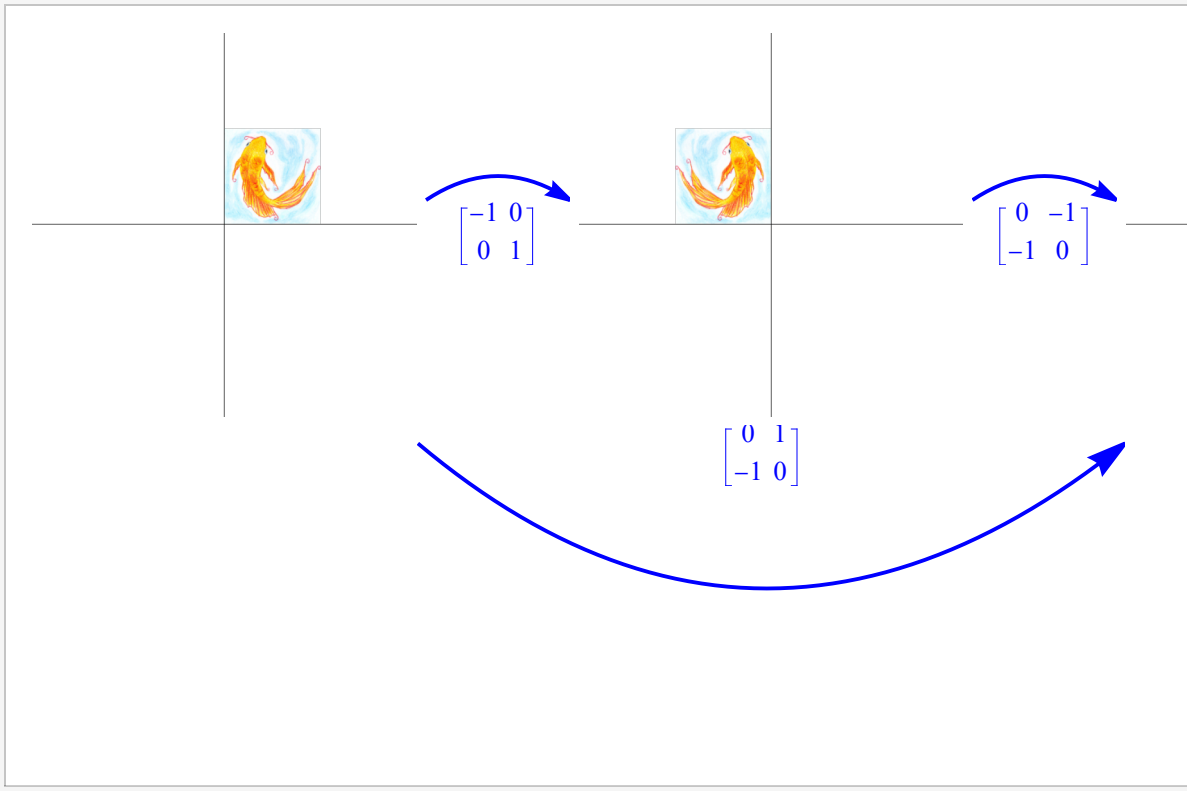


Remark:

Matrix multiplication is defined to make composition of functions (transformations) work.

Example:

Let $T : \mathbb{P}_3 \to M_{2\,x2}$ and $S : M_{2\,x2} \to \mathbb{R}^3$ so $ST : \mathbb{P}_3 \to \mathbb{R}^3$

$$T(f) = \begin{pmatrix} f(1) & f'(0) \\ 0 & f(1) + f''(1) \end{pmatrix}$$

$$S\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \begin{pmatrix} a \\ c \\ b + d \end{pmatrix}$$

Let $\beta = \{1, x, x^2, x^3\}$, $\gamma = \{E^{ij} \mid i, j \in \{1, 2\}\}$ and $\eta = \{e_i \mid i \in \{1, 2, 3\}\}$ be bases for the respective vector spaces above.

E^{ij} is the matrix with a 1 in row i and column j and zeros elsewhere.

e_i is the vector with a 1 in row i and zeros elsewhere.

To find $[T]_\beta^\gamma$, we should get a 4x4 matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 7 \end{bmatrix}$

$$T(1) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T(x) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = 1\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T(x^2) = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$T(x^3) = \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix}$$

$$[S]_\gamma^\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \quad S\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \quad S\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

To find $[ST]_\beta^\eta$, we should get a 3×4 matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 7 \end{bmatrix}$

$$ST(1) = S\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$ST(x) = S\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$ST(x^2) = S\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

$$ST\left(x^3\right) = S\begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}$$

$[S]_\gamma^\eta [T]_\beta^\gamma =$

In[ ]:= $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} . \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 7 \end{pmatrix}$

Out[ ]= $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 7 \end{pmatrix}$

An Aside to Look at Matrix Multiplication in More Detail

Definition:

Let A be an $m \times n$ matrix, let x be an $n \times 1$ vector, and let B be an $n \times p$ matrix, all with entries from the field $\mathbb{F}$. We define

$$Ax = \sum_{k=1}^{n} x_k\, a_k, \text{ where } x_k \text{ is the } k-\text{th entry in } x \text{ and } a_k \text{ is the } k-\text{th column of } A.$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix} = \begin{pmatrix} 10 \\ 28 \\ 46 \end{pmatrix}$$

We define

$AB = \left[Ab_1 \;\; Ab_2 \;\; \ldots Ab_p\right]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 2 & 1 \\ 1 & 1 \end{pmatrix} = \left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 10 & 4 \\ 28 & 7 \\ 46 & 10 \end{pmatrix}$$

Observation:

The columns of AB consist of linear combinations of the columns of A using the weights from each column from B.

FUNCTION NOTATION: $g\,(f\,(x))$ means apply f to x, then apply g to $f\,(x)$.

Theorem:

Let $T : U \to V$ and $S : V \to W$ be linear transformations on finite dimensional vector spaces, U, V, and W, over the same field $\mathbb{F}$. Let α, β, and γ be bases of each vector space respectively.

Set $B = [T]_\alpha^\beta$ and $A = [S]_\beta^\gamma$. Then $[ST]_\alpha^\gamma = AB$.

Proof:

Let $\{u_1, \ldots, u_q\}$, $\{v_1, \ldots v_r\}$, and $\{w_1, \ldots, w_s\}$ be basis for U, V, and W respectively.

To get the j-th column of $[ST]_\alpha^\gamma$, we write the vector $ST(u_j)$ in terms of $w_1, \ldots, w_s$. We can compute $ST(u_j)$ by applying S to the vector $T(u_j)$, where $T(u_j)$ is written in terms of $v_1, \ldots v_r$.

Notice that the coefficients of $T(u_j)$ written in terms of the β basis are exactly the numbers we put in column j of B. Thus $T(u_j) = b_{1j}\, v_1 + \ldots + b_{rj}\, v_r$.

Apply S to get $ST(u_j) = S(b_{1j}\, v_1 + \ldots + b_{rj}\, v_r) = b_{1j}\, S(v_1) + \ldots + b_{rj}\, S(v_r)$.

Notice that the coefficients of $S(v_k)$ written in terms of the γ basis are exactly the elements we put in column k of A. Thus $S(v_k) = a_{1k}\, w_1 + \ldots + a_{mk}\, w_m$.

Putting it all together:

$ST(u_j) = b_{1j}(a_{11}\, w_1 + \ldots + a_{s1}\, w_s) + \ldots + b_{rj}(a_{1r}\, w_1 + \ldots + a_{sr}\, w_s)$

$= (b_{1j}\, a_{11} + b_{2j}\, a_{12} + \ldots + b_{rj}\, a_{1r})\, w_1 + (b_{1j}\, a_{21} + b_{2j}\, a_{22} + \ldots + b_{rj}\, a_{2r})\, w_2 + \ldots + (b_{1j}\, a_{s1} + b_{2j}\, a_{s2} + \ldots + b_{rj}\, a_{sr})\, w_s$

So the j-th column of $[ST]_\alpha^\gamma$ looks like

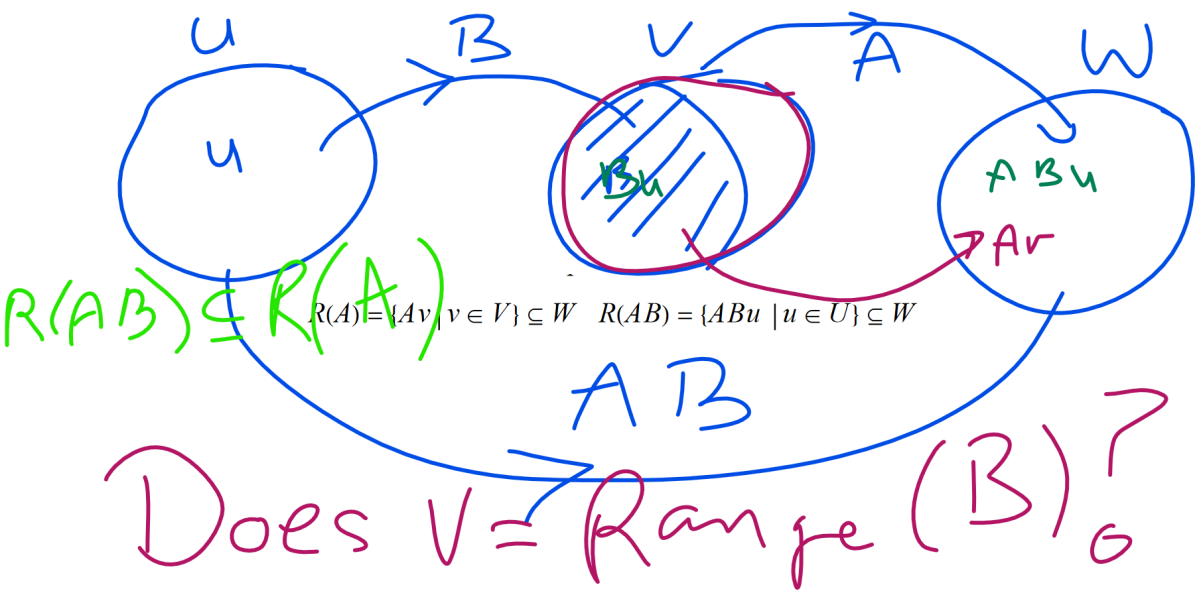
$$\begin{bmatrix} b_{1j}\, a_{11} + b_{2j}\, a_{12} + \ldots + b_{rj}\, a_{1r} \\ b_{1j}\, a_{21} + b_{2j}\, a_{22} + \ldots + b_{rj}\, a_{2r} \\ \vdots \\ b_{1j}\, a_{s1} + b_{2j}\, a_{s2} + \ldots + b_{rj}\, a_{sr} \end{bmatrix} = b_{1j} \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{s1} \end{bmatrix} + b_{2j} \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{s2} \end{bmatrix} + \ldots + b_{rj} \begin{bmatrix} a_{1r} \\ a_{2r} \\ \vdots \\ a_{sr} \end{bmatrix} = Ab_j$$

Question:

What is the relationship between $R(A)$ and $R(AB)$?

$R(A) = \{Av \mid v \in V\} \subseteq W \quad R(AB) = \{ABu \mid u \in U\} \subseteq W$

Notice that $R(AB) \subseteq R(A)$



What does your answer translate to in terms of linear transformations?

$R(ST) \subseteq R(S)$

Observation:

Let A be an $m \times n$ matrix and let B be an $n \times p$ matrix with entries from the field $\mathbb{F}$. Notice $C = AB$, where

$$c_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj} = [\text{Row}_i\, A][\text{Col}_j\, B]$$

viz. - it is the product of row i of A with column j of B.

$$AB = \begin{bmatrix} 3 & 5 & -1 & 2 \\ -2 & 1 & -4 & 1 \\ 0 & 3 & 0 & -4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 0 & -2 & 2 \\ 4 & 1 & 3 \\ 2 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 6 & -11 & 22 \\ -18 & -6 & -6 \\ -8 & -6 & -18 \end{bmatrix}$$

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$$(3)(2) + (5)(0) + (-1)\ (4) + (\ 2)\ (\ 2) = 6$$

entry in row 1 2 3

entry in column 1 2 3

Matrix multiplication.

Proof:

$C = AB$

$$\begin{aligned} c_{ij} = \mathrm{Row}_i[Ab_j] &= \mathrm{Row}_i[b_{1j}\, a_1 + b_{2j}\, a_2 + \ldots + b_{nj}\, a_n] \\ &= b_{1j}\, a_{i1} + b_{2j}\, a_{i2} + \ldots + b_{nj}\, a_{in} \\ &= \textstyle\sum_{k=1}^{n} a_{ik}\, b_{kj} = [\mathrm{Row}_i\, A][\mathrm{Col}_j\, B] \end{aligned}$$

Question:

If $AB = I$, what can you say about the relationship between the rows of A and the columns of B using the row-column product view of AB?

Observation:

Let A be an $m \times n$ matrix and let B be an $n \times p$ matrix with entries from the field $\mathbb{F}$. Then

$$AB = \textstyle\sum_{k=1}^{n} \mathrm{Col}_k[A]\, \mathrm{Row}_k[B]$$

Example:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 5 & 6 \\ 3 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 6 & 7 \\ 7 & 6 \\ 8 & 5 \\ 9 & 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} [\ 6\ 7\ ] + \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} [\ 7\ 6\ ] + \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix} [\ 8\ 5\ ] + \begin{bmatrix} 4 \\ 6 \\ 8 \end{bmatrix} [\ 9\ 4\ ]$$

$$= \begin{bmatrix} 6 & 7 \\ 12 & 14 \\ 18 & 21 \end{bmatrix} + \begin{bmatrix} 14 & 12 \\ 28 & 24 \\ 35 & 30 \end{bmatrix} + \begin{bmatrix} 24 & 15 \\ 40 & 25 \\ 56 & 35 \end{bmatrix} + \begin{bmatrix} 36 & 16 \\ 54 & 24 \\ 72 & 36 \end{bmatrix}$$

$$= \begin{bmatrix} 80 & 50 \\ 134 & 87 \\ 181 & 118 \end{bmatrix}$$

Proof:

$$c_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj} \;= \sum_{k=1}^{n} \mathrm{Row}_i[\mathrm{Col}_k(A)\,]\,\mathrm{Col}_j[\mathrm{Row}_k\, B] = \mathrm{Row}_i\,\mathrm{Col}_j\left[\sum_{k=1}^{n} \mathrm{Col}_k(A)\,\mathrm{Row}_k\, B\right]$$